

FEIT HACKATHON 2026 · RPM HIRE CHALLENGE

QueueGuard

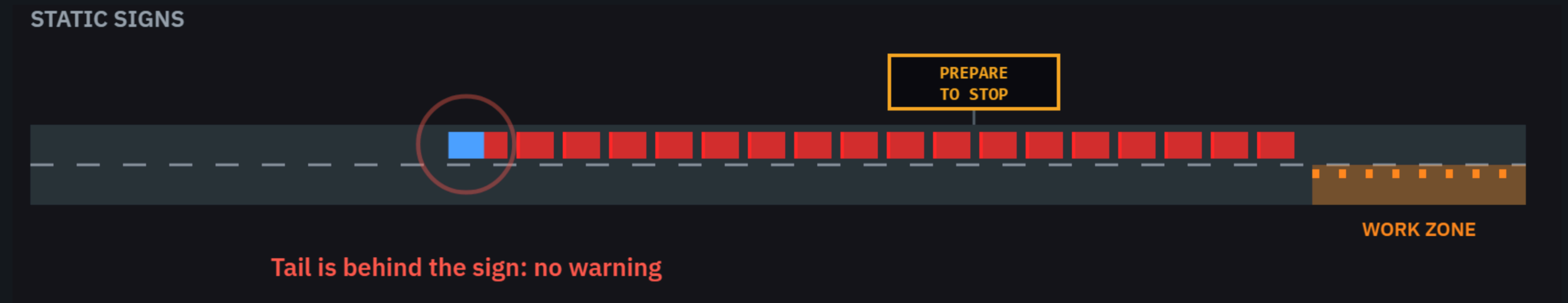
Warnings that move with the queue.

Team noname

End of Queue Detection

3 PM • A FREEWAY LANE CLOSURE

The queue moved. The signs didn't.



6 am

Signs set out before the merge

3 pm

The queue grows back past them

100 km/h

The next driver meets the tail
unwarned

HOW OFTEN DOES IT HAPPEN?

32%

of the time a queue exists, drivers get **no warning at a useful distance.**

Useful = 150–1,500 m before the tail: not too late, not already forgotten.

Warned in time · 68%

No timely warning · 32%

WHAT THE EVIDENCE SAYS

Queue warning works — when it is designed right.

I-35, TEXAS

up to -45%

work-zone crashes with an end-of-queue warning system

NSW FIELD TRIAL, 2025 · 13 SITES

Limited effect

from a commercial queue warning system and monochrome VMS

Detecting the queue is solved. A believable warning, in the right place at the right time, is not.

RPM HIRE ALREADY OWNS THE PARTS

The kit exists. The brain that links it doesn't.



VMS boards

the message



VSL trailers

the speed



Radar

the speeds



CCTV trailers

the evidence



RPM Track

the 4G link

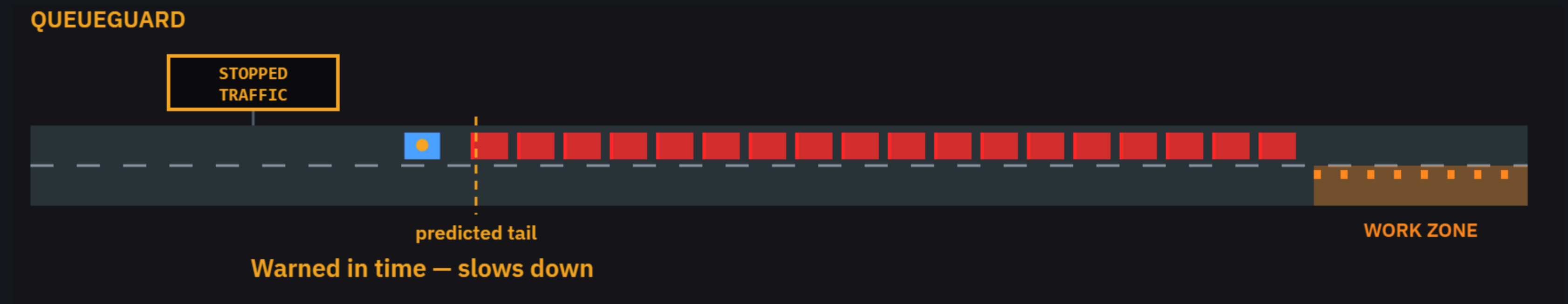


QueueGuard

One controller that links boards, speed limits and alerts, plus a site guideline — what the brief asks for.

HOW QUEUEGUARD WORKS

Sense. Predict. Act. Guide.



1 - Sense

Radar speeds every 20 s, every 400 m

2 - Predict

Where the queue tail will be in 90 s

3 - Act

Colour VMS at stopping distance · VSL 100 → 80 → 60
· worker and truck alerts

4 - Guide

A plan for each site, tested in a virtual trial

DEMO · THE WORKING PROTOTYPE

Same traffic. Same drivers. Only the warnings differ.



Top: static signs

Bottom: QueueGuard

Red ring: severe conflict



Run it yourself:
morlan1hp.github.io/queueguard

RESULTS · 40 SEEDS, ABOUT 43 SIMULATED PEAK HOURS PER OPTION

68% fewer severe end-of-queue conflicts.

Static signs



0.72 per hour

Fixed-position queue warning



0.67 · -6%

QueueGuard



0.23 · -68%

-14 km/h

arrival speed at the queue (85th
percentile 67.5 → 53.3)

32% → 0%

queue time with no timely warning

1,113 veh/h

throughput — unchanged, so safety
costs no capacity

ROBUST, BUILDABLE, SAFE

We tried to break it.

ROBUST

8 / 8

stress tests beat static signs:
–29% to –94% severe conflicts

Without speed limits 0.46, without
prediction 0.37, full system 0.23
per hour: both parts help.

BUILDABLE

0

new hardware: radar speeds in,
sign messages out, over RPM
Track

Controller and simulator run in a
web browser. Open source.

SAFE BY DESIGN

60 s

without data → every board
shows PREPARE TO STOP

A person confirms every truck
broadcast. Limits stay inside the
approved traffic plan.

WHAT IS NEW

Not just detect-and-warn: predict, place, link, prove.

	Typical queue warning	QueueGuard
Warning position	fixed boards	follows the predicted tail
Warning timing	wherever a board sits	150–1,500 m before the tail
VMS, VSL, alerts	separate products	one controller
False alarms	varies	never without a detected queue
Before set-up	standard layout	virtual trial of the site

Detection is the easy part. The value is **where, when and how believably** the warning appears.

VIABILITY

A software add-on to kit RPM already hires.

RPM Hire

A per-site-week licence on VMS, VSL and radar it already owns: more revenue per trailer, no new capital.

Contractors and agencies

Safety KPIs and an event log for their reporting, measured the way TfNSW measures.

Workers and drivers

Fewer crashes into the back of the queue — and into the crew.

US\$1.4–1.8M

societal crash costs saved (estimated) by the Texas I-35 queue warning system.
Uptake: works inside the approved traffic plan, supervisor in the loop.

Built now. Shadow mode next.

DONE

- Controller and traffic simulator
- 40-seed evaluation, 8 stress tests
- Site guideline and virtual trial
- Open source, live online

NEXT · MONTHS 1–2

- RPM Track radar feed, one live site
- Shadow mode: no control of signs
- Predicted tail checked against CCTV
- Thresholds tuned

THEN · MONTHS 3–4

- Live VMS and VSL control on 1–2 long-term sites
- TfNSW before–after evaluation
- Hire add-on for every site

NEXT STEP

One live site. No new hardware.

With the radar speeds RPM Track already logs, QueueGuard can run in shadow mode on a live RPM site within weeks — and prove its predictions against CCTV before it controls a single sign.

The queue moves. Now the warning does too.

Thank you · Team no name · morlan1hp.github.io/queueguard



APPENDIX · FOR QUESTIONS

The detail behind the numbers.

1. Full results: every option, every metric
2. Stress tests
3. Control logic in one page
4. What we don't know yet — and how the pilot finds out
5. Site guideline and virtual trial

Every option, every metric.

Option	Severe conflicts /h	Collisions /h	Arrival speed p85	Warned in time	Queue time unwarned	Throughput veh/h
Static signs	0.72 ± 0.33	0.23	67.5	78%	32%	1,113
Fixed-position queue warning	0.67 ± 0.30	0.25	65.8	83%	27%	1,112
QueueGuard	0.23 ± 0.17	0.07	53.3	100%	0%	1,113
QueueGuard without VSL	0.46 ± 0.29	0.14	64.7	100%	0%	1,112
QueueGuard without prediction	0.37 ± 0.27	0.05	55.1	100%	0%	1,113

Paired by seed (same drivers): QueueGuard – static = –0.48 per hour, 95% CI –0.85 to –0.12 (relative –68%, bootstrap 95% CI –29% to –90%). Fixed-position – static = –0.05, 95% CI –0.39 to +0.30: no measurable difference.

Mean ± 95% CI over 40 seeds (about 43 simulated hours per option), same arrivals and drivers for every option. Mean travel time 713–716 s for every option. Severe conflict = $TTC \leq 1$ s or $DRAC \geq 3.4$ m/s² (TfNSW / Deakin). Collisions are rare events: read them as a direction, not a precise rate.

Ahead of static signs in all 8 settings.

Setting (severe conflicts per hour)	Static signs	QueueGuard	Change
Base: 60% notice signs, queue spotted at 150–300 m, 1,800 veh/h	0.88	0.18	–79%
Only 20% of drivers notice a sign	0.92	0.60	–35%
40% notice a sign	1.11	0.37	–67%
80% notice a sign	0.32	0.23	–29%
Unwarned drivers spot the queue at 100–200 m	4.89	0.28	–94%
Unwarned drivers spot the queue at 250–450 m	0.97	0.14	–86%
Peak demand 1,600 veh/h	0.55	0.14	–75%
Peak demand 2,000 veh/h	1.43	0.46	–68%

Simple rules a traffic engineer can audit.

Detect	20 s mean speed below 35 km/h for two periods = queued; above 55 km/h to clear
Locate	Tail interpolated between radars 400 m apart
Predict	Tail speed fitted by least squares over the last 180 s, projected 90 s ahead
Place	First board at least 300 m upstream of the predicted tail: STOPPED TRAFFIC / PREPARE TO STOP, flashing; boards further upstream: QUEUE AHEAD x KM
Slow	80 within 2.1 km of the risk point, 60 within 900 m; neighbouring signs differ by at most 20 km/h; relax only after a 60 s hold
Alert	Crew alert when a vehicle passes a radar within 450 m of the tail above 80 km/h (dashboard now, phones in the pilot) · UHF CB 40 message drafted, a person confirms
Trust	Escalate at once, relax only after a 60 s hold; no queue message without a detected queue
Fail-safe	No data for 60 s: every board PREPARE TO STOP, VSL 80

What we don't know yet — and how the pilot finds out.

Open question	How we close it
Real drivers, not simulated ones	Shadow mode on RPM Track data, predicted tail against CCTV; then a TfNSW-style before–after pilot
How many drivers notice a sign (we assume 60%)	Stress-tested from 20% to 80%: QueueGuard stays ahead; the pilot measures real speed change
Tail prediction error $\approx$ 150 m (radar every 400 m, 20 s averaging)	No more precise than the current estimate; its value is moving warnings upstream sooner (late placements 10.9% $\rightarrow$ 6.7%). Denser radar or probe data can sharpen it
Legal use of variable limits	Only values inside the approved traffic plan and speed-zone authorisation; manual override of signs comes with the pilot build

Every site gets a plan — and a virtual trial.

QueueGuard guideline

Fill in the site. The end-of-queue plan on the right updates instantly; print it or save it as a PDF for the traffic management plan.

Site / road

Example freeway — outbound, lane closure for barrier

Posted speed (km/h)

100

Lanes in this direction

2

Lanes closed

1

Peak demand (veh/h, this direction)

1800

Work-zone capacity per open lane (veh/h)

1250

Heavy vehicles (%)

10

Time demand exceeds capacity (h)

0.5

Crest or curve limits sight of the queue

No

Works period

Night

Run virtual trial for this site (~20 s)

Print / save as PDF

[← Live demo](#) · [Evaluation](#)

1	Queue detected / cleared	Dashboard notification	Site supervisor
2	Tail growing > 1.8 km/h, or predicted to pass the most upstream active board within 5 min	Draft UHF CB 40 message → supervisor confirms (semi-automatic); deploy an extra VMS or a TMA upstream	Site supervisor, traffic controllers
3	Vehicle faster than 80 km/h within 450 m of the tail	Vibration alert on worker phones / wearables; log the event	All workers near the tail

UHF is semi-automatic on purpose: in the same field trial, manual UHF broadcasts reduced speeding while automatic ones showed no measurable effect.

6 · Monitoring and review

- Daily: queue minutes, maximum queue length, share of queue time with a timely warning (target 100%), false-warning share (target < 10%).
- Weekly: 85th-percentile arrival speed at the tail from radar; severe conflicts ($TTC \leq 1$ s or $DRAC \geq 3.4$ m/s²) from CCTV-trailer video, the surrogate safety measure used by TfNSW/Deakin.
- Escalate the plan if the tail passes the most upstream board on two shifts, or if severe conflicts exceed the baseline.

7 · Virtual trial

Measure (mean of 6 simulated hours)	Static signs	QueueGuard
Severe end-of-queue conflicts per hour ($TTC \leq 1$ s or $DRAC \geq 3.4$ m/s ²)	0.62	0.31
Arrival speed at the queue, 85th percentile (km/h)	67	54
Queue time with no timely warning	28%	0%
Throughput (veh/h)	1123	1120
Maximum queue length (m)	1892	1817

Expected change in severe end-of-queue conflicts at this site: **-50%**. Few seeds, so treat this as an indication; the full evaluation (eval.html) uses 40. Driver-behaviour assumptions are listed in docs/ASSUMPTIONS.md.

8 · Alignment

- Austrorads Guide to Temporary Traffic Management (AGTTM), incorporated in Victoria’s Code of Practice for Worksite Safety – Traffic Management (from 1 December 2023).
- Transport for NSW / Deakin University, *Work Zone End of Queue Study – Summary Report* (IMOVE, April 2025).
- Stopping sight distance per the Austrorads road design method (reaction time and deceleration coefficients to be confirmed by the designer).

Planning aid only.

Capacity, queue length and device positions are estimates from simple queuing and shockwave theory. A qualified traffic management designer must verify them against the site, the approved traffic management plan and the speed-zone authorisation before deployment.

In Speed, lanes, demand, capacity, heavy vehicles, sight lines

Out Queue length and growth, device positions, message library, thresholds, alert roles, KPIs

Virtual trial This site's speed, demand and truck share on our standard closure, static signs against QueueGuard, 6 seeds: –50% severe conflicts in this example

Use Printable for the traffic management plan; a planning aid a qualified designer verifies

A5